

Zero integral of a compactly supported divergence

`MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`
theorem

Makes all faces in a box divergence formula vanish by placing the support strictly inside the box.

Statement

Fix $n \geq 1$. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 and have compact topological support. Let $DF(x)$ denote the real Fréchet derivative and e_j the j th standard coordinate vector. Write

$$F_j(x) := (F(x))_j, \quad \partial_j F_j(x) := (DF(x)e_j)_j, \quad \operatorname{div} F(x) := \sum_{j=1}^n \partial_j F_j(x).$$

For Lebesgue measure on the coordinate space,

$$\int_{\mathbb{R}^n} \operatorname{div} F(x) dx = 0.$$

Assumptions

The support convention throughout this card is

$$\operatorname{supp} F := \overline{\{x \in \mathbb{R}^n : F(x) \neq 0\}}.$$

This closed set is assumed compact. The differentiability hypothesis is exactly C^1 . The domain has its coordinate sup norm. No externally specified integration domain or boundary-regularity assumption is needed.

Conclusion

The divergence is integrable and its whole-space integral vanishes. Compact support is imposed on F , which also forces the divergence to vanish off that support.

Notes

For $n = 1$ the two faces are endpoints and the same formula is the fundamental theorem of calculus. The source represents face coordinates by inserting one fixed coordinate into an $(n - 1)$ -tuple.

Proof route

Choose a box with the support strictly inside it. Write the box divergence formula as a sum of upper-face minus lower-face integrals. Each face value is zero, and the divergence is zero outside the box, so the whole-space integral is zero.

▼ Proof steps

1. **Enclose the support and justify integrability.** Choose $R > 0$ such that

$$\text{supp } F \subset (-R, R)^n, \quad Q = [-R, R]^n.$$

If $x \notin \text{supp } F$, some neighborhood of x is disjoint from the nonzero set of F . On that neighborhood $F = 0$, and hence $DF(x) = 0$. Therefore

$$x \notin \text{supp } F \implies \partial_j F_j(x) = 0 \text{ for every } j \implies \text{div } F(x) = 0.$$

The divergence is continuous because F is C^1 , so it is integrable on compact Q . It is zero on Q^c , and is therefore integrable on $\mathbb{R}^n$ as well.

2. **Spell out the face formula.** Fix i . Let $Q_i = \prod_{k \neq i} [-R, R]$ and write $y = (y_k)_{k \neq i}$. Insert the missing coordinate by

$$(\iota_i(t, y))_k = \begin{cases} t, & k = i, \\ y_k, & k \neq i. \end{cases}$$

For fixed y , the scalar function $t \mapsto F_i(\iota_i(t, y))$ is C^1 and

$$\frac{d}{dt} F_i(\iota_i(t, y)) = (DF(\iota_i(t, y))e_i)_i = \partial_i F_i(\iota_i(t, y)).$$

The face expression in the box divergence theorem is precisely the following repeated-integral calculation (Fubini on the compact box, followed by the one-dimensional fundamental theorem of calculus):

$$\begin{aligned} \int_Q \partial_i F_i(x) dx &= \int_{Q_i} \left[\int_{-R}^R \frac{d}{dt} F_i(\iota_i(t, y)) dt \right] dy \\ &= \int_{Q_i} [F_i(\iota_i(R, y)) - F_i(\iota_i(-R, y))] dy. \end{aligned}$$

Sum over i to obtain $\int_Q \text{div } F$. In the source this sum is obtained directly from the cited box divergence theorem: C^1 supplies continuity and a derivative everywhere, the exceptional set is empty, and Step 1 supplies integrability. The calculation above explains its face terms.

3. **Set the face values to zero and pass to the whole space.** The i th coordinate of $\iota_i(\pm R, y)$ is $\pm R$. Hence

$$\iota_i(\pm R, y) \notin (-R, R)^n \implies \iota_i(\pm R, y) \notin \text{supp } F \implies F_i(\iota_i(\pm R, y)) = 0.$$

Each integrand in the last line of Step 2 is thus $0 - 0$. Since Step 1 also gives $\text{div } F = 0$ on Q^c ,

$$\begin{aligned} \int_{\mathbb{R}^n} \text{div } F &= \int_Q \text{div } F + \int_{Q^c} \text{div } F \\ &= \sum_{i=1}^n \int_{Q_i} (0 - 0) dy + 0 = 0. \end{aligned}$$

For $n = 1$, Q_i is the zero-dimensional one-point product; the formula is simply the difference of the two endpoint values.

Main citations

- Exact declaration and its source —
MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport
- Box divergence formula with a countable exceptional set —
MeasureTheory.integral_divergence_of_hasFDerivAt_off_countable
- Coordinate definition of divergence — MathlibAnnex.Piola.coordinateDivergence
- Divergence vanishes off the topological support —
MathlibAnnex.Piola.coordinateDivergence_eq_zero_of_not_mem_tsupport
- An open box enclosing a compact set —
MathlibAnnex.Piola.exists_open_box_containing_compact

Lean source signature (exact)

```
theorem integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport
  {m : ℕ} {F : (Fin (m + 1) → ℝ) → (Fin (m + 1) → ℝ)}
  (hF : ContDiff ℝ 1 F) (hFc : HasCompactSupport F) :
  ∫ x, coordinateDivergence F x = 0
```

In the source	Mathematical meaning
$\{m : \mathbb{N}\} \{F : (\text{Fin } (m + 1) \rightarrow \mathbb{R}) \rightarrow (\text{Fin } (m + 1) \rightarrow \mathbb{R})\}$	The positive dimension $n = m + 1$ and vector field $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$.
$(hF : \text{ContDiff } \mathbb{R} \ 1 \ F)$	The exact regularity assumption is C^1 .
$(hFc : \text{HasCompactSupport } F)$	The closed topological support $\text{supp } F = \overline{\{x : F(x) \neq 0\}}$ is compact.

In the source	Mathematical meaning
$\text{coordinateDivergence } F \ x$	The function $\text{div } F(x) = \sum_j (DF(x)e_j)_j = \sum_j \partial_j F_j(x)$, where $F_j = (F)_j$.
$\int x, \text{coordinateDivergence } F \ x = 0$	The whole-space Lebesgue integral $\int_{\mathbb{R}^n} \text{div } F(x) \, dx = 0$. The theorem is not merely a box integral identity.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators Topology

namespace MathlibAnnex
namespace Piola
```

▼ **Exact content identity**

Declaration: `MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`

Accepted content SHA-256: 6615269efd172654102c91e02a3056f51f399018a4faa42e1acd7c85d2b63488

Accepted source guide SHA-256: 9284ee529653d2155bcbe3ebf7302736c29081fc1b35539edec2dc2ff6c5bed0

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73